

Concurrency and Distributed Systems

Week 6 – Part 2: Granularity and Break-Even

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Part 2 Roadmap

- 1 Granularity
- 2 Overhead
- 3 Measurement
- 4 Break-Even
- 5 Exercise 2

The Main Tradeoff

Parallelism is not free

Fork/Join adds splitting, scheduling, synchronization, and joining. The computation must be large enough to repay those costs.

- Small inputs often lose.
- Medium inputs may be noisy.
- Large independent inputs are the best candidates.

Granularity

- Granularity means how much work each task performs before splitting stops.
- The threshold is the control knob.
- A threshold of one element usually creates too many tasks.
- A threshold larger than the input creates no parallelism.

Where Overhead Comes From

- Allocating task objects.
- Pushing tasks into worker dequeues.
- Stealing tasks between workers.
- Joining and combining results.
- Cache effects and memory bandwidth.

Measurement warning

Faster on one laptop does not mean faster everywhere.

Measurement Probe

```
1 long start = System.nanoTime();  
2 long sequential = sequentialSum(values);  
3 long sequentialNanos = System.nanoTime() - start;  
4  
5 start = System.nanoTime();  
6 long parallel = forkJoinSum(values, threshold);  
7 long forkJoinNanos = System.nanoTime() - start;
```

Measure the same input and check that both implementations return the same result.

A Defensible Measurement

- Use the same data for sequential and Fork/Join.
- Warm up before trusting numbers.
- Repeat runs and record variation.
- Report hardware, JDK, input sizes, and threshold.
- Explain the result instead of only listing times.

Break-Even Point

Definition

The break-even point is the smallest tested input size where Fork/Join is faster than the sequential baseline.

- It may not appear in the tested range.
- It can move when threshold, CPU, or JVM changes.
- It is evidence, not a universal law.

Demo04 and Demo05

- Demo04_BreakEvenProbe: compares sequential and Fork/Join timing.
- Demo05_GranularityCost: estimates leaf task count from size and threshold.

Class discussion

If threshold goes down, what happens to task count? What should happen to overhead?

Exercise 2 Contract

- Preserve the input size order in the report.
- Return one row per input size.
- Check that sequential and Fork/Join results match.
- Return the first size where Fork/Join wins, or -1 .
- Explanation must mention overhead.

Exit Check

- ① What happens if the threshold is too small?
- ② Why must the report include correctness checks?
- ③ Why is a break-even point machine-dependent?